Problem Set # 2: Understanding Factors Affecting the Early Care and Education Workforce in the United States

The mental health of early childhood educators is a critical component of a high-quality early care and education system (Murphy & Reid, 2025). Research has shown that educators' psychological well-being affects not only their own work experiences but also children's classroom environments, emotional development, and academic achievement (McLean & Connor, 2015; Oberle & Schonert-Reichl, 2016). In addition, educators with better mental health tend to have more positive interactions with children and report fewer behavioral challenges in their classrooms (Jeon et al., 2014; Kwon et al., 2019). Because early childhood experiences have lasting effects on children's health, educational attainment, and future opportunities, understanding the factors associated with educator mental health is an important research and policy priority (Murphy & Reid, 2025).

In this problem set, you will examine factors associated with depression among staff working in center-based early care and education programs in the United States. Following the analytical approach of Murphy and Reid (2025), you will use data from the 2019 National Survey of Early Care and Education (NSECE) to apply the statistical techniques covered in class and investigate how educator characteristics are associated with depression.

1. **Clean and understand the data.** Subset the NSECE dataset to include the following variables: **(4 points)**
    - WF9_WORK_WAGE (hourly wage)
    - WF9_WORK_ROLE (staff role)
    - WF9_CHAR_RACE (race)
    - WF9_CHAR_EDUC (highest level of education completed)
    - WF9_CHAR_YEAR_BORN (staff's birth year)
    - WF9_CESD7_TOT (staff's total score on CESD7 (depression) scale)[1]

Rename the variables using clear, descriptive names. Perform any data cleaning you consider necessary. Create the following variables using the following code:

[1] This variable measures the respondent's overall level of depressive symptoms. Scores range from 0, indicating no signs of depression, to 21, indicating severe depressive symptoms. According to the NSECE User Guide (2019), a score of 8 or higher is the recommended cutoff for identifying individuals who may have major depressive disorder. For additional information about this measure, see page 220 of the NSECE User Guide (2019).

```
[name of your new dataset here] <-
      [name of your previous dataset here] %>%
      mutate(staff_age = 2019 - WF9_CHAR_YEAR_BORN,
             staff_age_new = case_when(
                  staff_age >=20 & staff_age <=40 ~ "Between 20 and 40",
                  staff_age >=41 & staff_age <=50 ~ "Between 41 and 50",
                  staff_age >= 51 ~ "More than 51"),
             staff_age_new = factor(staff_age_new,
                                    levels = c("Between 20 and 40",
                                               "Between 41 and 50",
                                               "More than 51"),
                                    labels = c("Between 20 and 40",
                                               "Between 41 and 50",
                                               "More than 51")),
            hr_wage_new = case_when(
                  hr_wage <=10 ~ "Less than $10",
                  hr_wage >=11 & hr_wage <=20 ~ "Between $11 and $20",
                  hr_wage >=21 & hr_wage <=30 ~ "Between $21 and $30",
                  hr_wage >=31 ~ "More than $31"),
            hr_wage_new = factor(hr_wage_new,
                              levels = c("Less than $10",
                                         "Between $11 and $20",
                                         "Between $21 and $30",
                                         "More than $31"),
                              labels = c("Less than $10",
                                         "Between $11 and $20",
                                         "Between $21 and $30",
                                         "More than $31")))
```

Using the variables role, race, educational attainment, depression, and the new staff age and hourly wage variables (staff_age_new and hr_wage_new), complete the following tasks for each variable:

- Identify its level of measurement (nominal, ordinal, or interval/ratio).
- Indicate whether it is discrete or continuous.
- Present the appropriate descriptive statistics, tables, or graphs, including measures of central tendency, dispersion, and shape when applicable.
- Describe and interpret your findings.

[note: full visualizations using ggplot2 are *not* required].

2. **Prepare the data for linear regression.** Create the derived variables needed to estimate a multiple linear regression model: **(2 points)**

   - Create k-1 binary variables for staff role (k = number of categories). After examining the variable, choose a reference group.
   - Create k-1 binary variables for race (k = number of categories). After examining the variable, choose a reference group.
   - Create k-1 binary variables for educational attainment (k = number of categories). After examining the variable, choose a reference group.

- Create k-1 binary variables for staff age (k = number of categories). After examining the variable, choose a reference group.
- Create k-1 binary variables for hourly wage (k = numbeer of categories). After examining the variable, choose a reference group.

3. **Estimate the OLS regression model.** Estimate the following OLS regression model to examine factors associated with depression among center-based early care and education staff: **(4 points)**

$$\text{Depression} = \beta_0 + \beta_{1..}\beta_3(\text{hourly wage}) + \beta_{4..}\beta_6(\text{staff role}) + \beta_{4..}\beta_6(\text{staff age}) + \beta_{34\ldots}\beta_{11}(\text{Education Level categories}) + \beta_{12\ldots}\beta_{14}(\text{Race categories}) + \varepsilon$$

Answer the following questions:

- Interpret the coefficient estimates for each statistically significant independent variable.
- Interpret the overall model fit (e.g., R^2 and adjusted R^2).
- What conclusions can you draw about the factors associated with depression among center-based early care and education staff?
- Create a residual plot and a histogram of the model residuals. Do these diagnostic plots suggest that the assumptions of linearity, homoscedasticity, and normality are reasonably met? What hypotheses do you have about why the plots look the way they do?